

Assignment 3 (Lab Report)

Basic Bioinformatics (BM2023)

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Single-Letter Codes That Do Not Represent Standard Amino Acids

The 20 standard amino acids are represented by the following single-letter codes:

A, R, N, D, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y, V

Therefore, the letters that **do not** represent any of the 20 standard amino acids are:

<i>B, J, O, U, X, Z</i>

Explanation

Some of these letters are used in special or ambiguous contexts:

- **B** – Represents either Aspartate (D) or Asparagine (N) (ambiguous).
- **Z** – Represents either Glutamate (E) or Glutamine (Q) (ambiguous).
- **X** – Unknown or unspecified amino acid.
- **J** – Sometimes used to denote Leucine (L) or Isoleucine (I) (non-standard).
- **O** – Pyrrolysine (rare, not among the standard 20 amino acids).
- **U** – Selenocysteine (a special case amino acid).

Important Note

The acidic, basic, and aromatic amino acids are valid single-letter codes:

- **Acidic:** D (Aspartate), E (Glutamate)
- **Basic:** K (Lysine), R (Arginine), H (Histidine)
- **Aromatic:** F (Phenylalanine), W (Tryptophan), Y (Tyrosine)

All of the above are standard amino acids.

Cell Attachment Peptide Sequences

Cell attachment peptide sequences are short amino acid motifs present in extracellular matrix (ECM) proteins that facilitate adhesion of cells to the ECM. These sequences primarily interact with integrins and other cell surface receptors.

Common cell attachment motifs include:

- **RGD (Arg-Gly-Asp)** – Found in fibronectin and vitronectin; binds integrins.
- **DGEA (Asp-Gly-Glu-Ala)** – Present in collagen.
- **YIGSR (Tyr-Ile-Gly-Ser-Arg)** – Found in laminin.
- **IKVAV (Ile-Lys-Val-Ala-Val)** – Laminin-derived motif important in neuronal adhesion.

These motifs are responsible for cell adhesion, migration, proliferation, and differentiation.

Common Extracellular Matrix (ECM) Proteins

Major ECM proteins include:

1. **Fibronectin** – Glycoprotein containing RGD motif; promotes adhesion and wound healing.
2. **Collagen** – Most abundant ECM protein; forms triple helix structure.
3. **Laminin** – Major component of basement membrane.
4. **Integrin $\alpha 2b$** – Platelet receptor involved in clotting.
5. **Collagen Type II** – Important for cartilage structure.

BLAST Analysis of Peptide Sequences

The peptide sequences RGD, DGEA, YIGSR, and IKVAV were subjected to BLAST analysis against known ECM protein sequences.

Observations:

- Short peptide motifs do not always show strong sequence similarity due to their small size.
- YIGSR and IKVAV motifs are specifically found in laminin sequences.
- Collagen shows conserved Gly-X-Y repeats.

These motifs function as localized binding regions rather than conserved long domains.

Table 1: Common Regions Identified in Extracellular Matrix Proteins After BLAST Analysis

ECM Protein	Common/Conserved Region	Function
Collagen	Gly-X-Y repeating sequence (Glycine-rich region)	Forms triple helix providing tensile strength
Collagen Type II	Triple helix collagen domain	Structural support and cartilage stability
Fibronectin	RGD (Arg-Gly-Asp) motif	Integrin binding and cell adhesion
Laminin	IKVAV adhesion sequence	Cell attachment and basement membrane interaction
Integrin $\alpha 2b$	ECM-binding interaction region	Cell-ECM adhesion and signal transmission

Peptide Sequences Used for BLAST Analysis

- **Collagen**
GLPGPPGPPGPPGPPGPP
- **Collagen II**
GEPGPPGPPGPPGPPGPP
- **Fibronectin**
GRGDSPASSK

- **Laminin**
IKVAV
- **Integrin_alpha2b**
KVGFFKRNRPPEEDDEEG

Laminin Structure and Domain Architecture

Laminin is a large heterotrimeric glycoprotein composed of three chains:

- α chain
- β chain
- γ chain

It forms a characteristic cross-shaped structure in basement membranes.

Overall Structure

- One long arm formed by a triple coiled-coil structure.
- Three short arms from N-terminal regions.
- Globular domains at the C-terminal end of the α chain.

Structural Domains

1. **LN Domain** – Laminin N-terminal domain; mediates polymerization.
2. **LE Domains** – EGF-like cysteine-rich repeats.
3. **Coiled-Coil Domain** – Stabilizes heterotrimer assembly.
4. **Laminin G-like (LG) Domains** – Present at C-terminus of α chain; responsible for receptor binding.

Structural Folds Present

- Coiled-coil fold
- Beta-sandwich fold (LG domains)
- EGF-like fold

The LG domains contain calcium-binding sites and mediate interactions with integrins such as $\alpha6\beta1$.

Interesting Structural Feature

Laminin's cross-shaped structure allows it to act as a scaffold in basement membranes. The long coiled-coil arm provides mechanical stability, while the short arms enable network formation. The multidomain architecture makes laminin structurally complex compared to many other ECM proteins.

Human Insulin vs Pig Insulin BLAST Comparison

BLAST alignment of human insulin and pig insulin shows high sequence similarity.

Difference observed:

- At position B30:
 - Human insulin: Threonine (Thr)
 - Pig insulin: Alanine (Ala)

Thus, human and pig insulin differ by only one amino acid in the mature insulin sequence.

Conclusion

This study analyzed ECM attachment motifs, ECM protein classes, BLAST results, and the detailed structure of laminin. Laminin exhibits a complex multidomain structure with coiled-coil and LG domains essential for basement membrane integrity and cell adhesion. Comparative analysis of insulin sequences demonstrates evolutionary conservation with minimal amino acid variation.